# Longaeva Hackathon — Public Datasets

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# 1. Cross-Sector Alt Data

## 1.1 Consumer Transaction, Attention & Demand Proxies

| Dataset | Use Cases |
|---|---|
| **TSA Checkpoint Travel Numbers** | Daily traveler throughput; airline / leisure demand |
| **Google Trends** | Search intent, brand interest, event attention |
| **Apple App Store / Google Play rankings** | App momentum, consumer/mobile demand |
| **Wikimedia Pageviews API** | Attention proxy for tickers, brands, topics |
| **Reddit / Stocktwits public/community feeds** | Retail sentiment, theme detection |

## 1.2 Geospatial, Satellite & Mobility

| Dataset | Use Cases |
|---|---|
| **Copernicus Data Space Ecosystem** | Sentinel-1 / Sentinel-2 imagery for crops, construction, parking lots, storage, land use |
| **Landsat / EarthExplorer** | 30m imagery, long-history industrial / land-use analysis |
| **NASA FIRMS** | Fire and thermal anomaly monitoring; supply-chain and weather-shock analysis |
| **OpenStreetMap / Overpass API** | POIs, roads, buildings, retail footprint mapping |
| **NOAA MarineCadastre AIS** | Historical vessel tracking in US waters; port activity, congestion, shipping studies |
| **AISHub** | Near-real-time terrestrial AIS |
| **OpenSky Network** | ADS-B flight tracking, cargo flows, aviation activity |

## 1.3 Weather & Climate

| Dataset | Use Cases |
|---|---|
| **ERA5 Reanalysis / CDS** | Power demand, ag yields, weather normalization, retail traffic |
| **NOAA GFS / HRRR** | Short-term forecasts; energy and commodity modeling |
| **NOAA Climate Data Online** | Station-level weather history, degree days |

## 1.4 Labor, Hiring & Human Capital

| Dataset | Use Cases |
|---|---|
| **BLS (JOLTS, CES, CPS)** | Labor demand, payrolls, unemployment, wage inflation |

| Dataset | Use Cases |
|---|---|
| **Indeed Hiring Lab** | Sector hiring trends, labor demand signals |
| **H-1B LCA Disclosures** | Employer hiring intensity, skill demand, geographic talent maps |

## 1.5 Trade & Supply Chain

| Dataset | Use Cases |
|---|---|
| **US Census USA Trade Online** | HS-code imports/exports, port analysis |
| **UN Comtrade (Comtrade+)** | Global bilateral trade flows |
| **Eurostat Comext** | EU bilateral trade by product / geography |
| **Port of LA / Long Beach dashboards** | TEU throughput, import cycle bellwether |
| **Panama Canal Authority** | Transit counts, drought bottlenecks, chokepoint analysis |
| **IMF PortWatch** | Chokepoint stress, AIS-derived delay tracking |
| **FreightWaves SONAR (public charts)** | Trucking rates, tender volumes |

## 1.6 Energy & Commodities

| Dataset | Use Cases |
|---|---|
| **EIA Open Data** | Oil, gas, power inventories, generation, prices |
| **ENTSO-E Transparency Platform** | EU generation, load, interconnection flows |
| **USDA NASS / WASDE** | Crop production, acreage, inventory and forecast work |
| **Baker Hughes Rig Count** | Drilling activity |

## 1.7 Patents, Innovation & R&D

| Dataset | Use Cases |
|---|---|
| **USPTO Open Data Portal / PatentsView migration target** | Patents, assignees, citations, IP networks |
| **OpenAlex** | Scholarly works, institutions, citation graph |
| **NIH RePORTER** | Grants, research funding, academic / biotech pipeline themes |

## 1.8 Government, Politics & Public Spending

| Dataset | Use Cases |
|---|---|
| **FEC OpenFEC** | Campaign finance, committee flows, political donation analytics |
| **Lobbying Disclosure / LDA.gov** | Lobbying spend by issue / client / firm |
| **Capitol Trades** | Congressional trading activity |
| **USASpending.gov** | Federal awards, grants, contract analysis |
| **Federal Register** | Rulemaking, executive orders, agency actions |

## 1.9 News, Text & Sentiment

| Dataset | Use Cases |
|---|---|
| **GDELT 2.0** | Global news events, tone, themes, geocoded media monitoring |
| **Common Crawl** | Web-scale text corpora, snapshots, embeddings |
| **Financial Phrasebank** | Finance NLP benchmark / sentiment classification |

| Dataset | Use Cases |
| --- | --- |
| **SEC EDGAR Full-Text Search / data.sec.gov** | Management commentary, filing text change analysis |

# 2. Macro & Economic

| Dataset | Use Cases |
| --- | --- |
| **FRED** | Macro time series, rates, inflation, labor, credit |
| **World Bank Open Data** | Cross-country macro and development indicators |
| **IMF Data** | WEO, IFS, BOP, SDMX macro feeds |
| **ECB Data Portal** | Euro-area monetary and financial stability data |
| **BIS Data Portal** | Global banking, credit, debt, FX and reserve statistics |
| **NY Fed GSCPI / FRED `GSCPI`** | Supply-chain pressure composite |
| **NY Fed Nowcast** | Weekly GDP nowcast |
| **Cleveland Fed Inflation Nowcast** | Daily CPI / PCE nowcast |

# 3. Filings & Regulatory (Traditional)

| Dataset | Use Cases |
| --- | --- |
| **SEC EDGAR / Search Filings** | 10-K, 10-Q, 8-K, S-1, 20-F, text and XBRL |
| **13F Holdings via EDGAR / WhaleWisdom** | Institutional ownership, fund crowding, smart-money tracking |
| **N-PORT / N-CEN** | Fund holdings, ETF and fund exposure analysis |
| **Form 4 / Insider Transactions** | Insider buying and selling signals |

| Dataset | Use Cases |
|---|---|
| TRACE / FINRA | Corporate bond trading |
| OpenFIGI | Identifier mapping |
| Ken French Data Library | Fama-French and related factor series |
| Quiver Quantitative | Congress trades, lobbying, insider, contracts |
| SOV.AI | Alt-data bundles and factors |

# 4. Healthcare — FDA, CMS, NIH & CDC

## 4.1 FDA — Adverse Events & Safety Surveillance

| Dataset | Use Cases |
|---|---|
| FDA AEMS Public Dashboard | Unified adverse-event monitoring across multiple product classes |
| openFDA Drug Event API (FAERS) | Drug / biologic safety signals |
| MAUDE text search | Device adverse events search |
| openFDA Device Event API | Device event / malfunction monitoring |
| VAERS | Vaccine safety monitoring |
| FDA Recalls & Safety Alerts | Event-driven alerts |

## 4.2 FDA — Approvals, Labels & Competitive Landscape

| Dataset | Use Cases |
|---|---|
| Drugs@FDA | NDA/BLA approvals, labels, history |

| Dataset | Use Cases |
|---|---|
| **Orange Book** | Generic equivalence, exclusivity, LoE modeling |
| **Purple Book** | Biosimilar / interchangeable competition |
| **510(k) Database** | Device clearances |
| **PMA Database** | Class III approvals |
| **Breakthrough / Fast Track / Priority Review lists** | Pipeline acceleration monitoring |
| **Warning Letters** | Manufacturing / compliance risk |
| **DailyMed** | Machine-readable labels |

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## 4.3 Clinical Research Pipeline

| Dataset | Use Cases |
|---|---|
| **ClinicalTrials.gov API v2** | Trial starts, enrollment, completions, catalyst calendars |
| **EU Clinical Trials Register / CTIS** | European trial pipeline |
| **WHO ICTRP** | Global registry aggregation |

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## 4.4 CMS — Medicare, Medicaid & Provider Data

| Dataset | Use Cases |
|---|---|
| **Medicare Part D Spending by Drug** | Brand-level drug spend and claim counts |
| **Medicare Part B Spending by Drug** | Physician-administered drug spend |
| **CMS Drug Spending** | Medicaid / broader spending views |
| **Provider Utilization & Payment Data** | Procedure / provider volume trends |
| **Open Payments** | Manufacturer-to-physician payments |

| Dataset | Use Cases |
|---|---|
| Care Compare | Quality, hospitals, post-acute providers |
| MA Enrollment & Landscape Files | MA penetration and plan share |
| National Health Expenditure Accounts | Macro healthcare spending |

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## 4.5 NIH, CDC & HHS

| Dataset | Use Cases |
|---|---|
| CDC WONDER | Mortality, natality, public health time series |
| CDC FluView / RSV-NET / NREVSS | Respiratory season tracking |
| CDC NHANES / BRFSS | Prevalence, population health, TAM work |
| HCUP | Hospital admissions / ED / inpatient analysis |
| HRSA Area Health Resources File | County provider density and healthcare access |

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# 5. Industrials — Aviation, Shipping, Rail, Trucking

## 5.1 Aviation — Flight Tracking & Operations

| Dataset | Use Cases |
|---|---|
| ADS-B Exchange | Unfiltered flights, aircraft movement |
| FlightAware AeroAPI | Tail-level flight tracking, schedules, history |
| Eurocontrol "Our Data" | European traffic, delays, system-level operations |
| FAA OPSNET / ASPM | US airport ops counts, air traffic trends |

| Dataset | Use Cases |
|---|---|
| **BTS TranStats** | T-100, DB1B, Form 41, OTP, air carrier employment |
| **FAA Aircraft Registry** | Fleet age, ownership, tail mapping |
| **NTSB CAROL / accident data** | Aviation safety events |

## 5.2 Maritime — AIS, Ports & Shipping

| Dataset | Use Cases |
|---|---|
| **aisstream.io** | Live AIS websocket for prototyping |
| **Global Fishing Watch** | Vessel presence, sanctions / dark-fleet analysis |
| **OECD Maritime Trade Dashboard** | Pre-computed port / commodity aggregates |
| **BTS Port Performance Freight Statistics** | US ports, dwell, wait-time metrics |
| **Port of LA / Long Beach TEUs** | Container flow bellwethers |
| **UNCTADstat** | Global container throughput |

## 5.3 Shipping Rates & Indices

| Dataset | Use Cases |
|---|---|
| **Drewry World Container Index** | Container spot rates |
| **Freightos FBX** | Container rate index |
| **Cass Freight Index** | Shipments and expenditures index |

## 5.4 Rail

| Dataset | Use Cases |
|---|---|
| **AAR Data Center** | Weekly rail traffic |
| **STB Rail Service Data** | Train speed, dwell, cars on line |
| **STB Waybill (public-use subset)** | Commodity and routing sample |
| **STB R-1 reports** | Detailed financial filings |
| **FRA safety data** | Derailments and accidents |
| **USDA AgTransport Open Data** | Ag rail, barge, truck movement |
| **FRED rail series** | Cleaned rail aggregates |

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## 5.5 Trucking & Logistics

| Dataset | Use Cases |
|---|---|
| **FMCSA A&I / SAFER ecosystem** | Carrier census, safety, crash and inspection stats |
| **DAT Trendlines** | National van / reefer / flatbed rates |
| **EIA Weekly Diesel** | Fuel-cost pass-through |

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# 6. Financials

## 6.1 Banking

| Dataset | Use Cases |
|---|---|
| **FFIEC Call Reports** | Bank balance sheet, income, loan mix |
| **FDIC BankFind Suite API** | Institution data, historical financials, failures |
| **FDIC Summary of Deposits** | Branch deposit market share |

| Dataset | Use Cases |
| --- | --- |
| **UBPR** | Peer ratios and bank benchmarking |
| **NCUA call reports** | Credit union trends |
| **Fed H.8** | Weekly bank balance sheet aggregate |
| **Fed SLOOS** | Lending standards and demand |

## 6.2 Insurance

| Dataset | Use Cases |
| --- | --- |
| **NAIC Financial Data Repository** | Statutory filings and reserve work |
| **State DOI rate filings** | Pricing approvals, regulatory risk |
| **OpenFEMA NFIP datasets** | Flood policies, claims, penetration |
| **PCS CAT summaries** | Cat event summaries |

## 6.3 Consumer Credit & Household Finance

| Dataset | Use Cases |
| --- | --- |
| **Fed G.19 Consumer Credit** | Revolving / non-revolving credit cycle |
| **NY Fed Household Debt & Credit** | Household leverage and delinquencies |
| **NY Fed / Equifax CCP summaries** | Consumer credit trends |
| **CFPB Consumer Complaint Database** | Complaint sentiment, issuer risk proxy |
| **Fed Z.1 Financial Accounts** | Flow of funds, household balance sheets |

## 6.4 Mortgage & Housing Finance

| Dataset | Use Cases |
|---|---|
| **HMDA / CFPB platform** | Loan-level origination, denials, pricing |
| **Freddie Mac Single-Family Loan-Level Dataset** | Delinquency, prepay, loan performance |
| **Fannie Mae Single-Family Loan Performance** | Mortgage performance, prepay, CRT-style analysis |
| **Census New Residential Construction** | Starts, permits, completions |
| **MBA Mortgage Applications (summary)** | Purchase / refi demand |

## 6.5 Payments & Fintech

| Dataset | Use Cases |
|---|---|
| **Fed Payments Study** | Payment rails, cards vs ACH, checks |
| **Nacha ACH stats** | ACH growth, RTP adoption |
| **CFPB Supervisory Highlights** | Regulatory and supervisory pressure |
| **DeFi Llama** | TVL, fees, yields, protocol metrics |
| **CoinMetrics (free tier)** | On-chain metrics |

# 7. Consumer

## 7.1 Retail Sales & Consumer Spending

| Dataset | Use Cases |
| --- | --- |
| **Census MRTS** | Monthly retail sales |
| **BEA PCE** | Spending by category |
| **Census e-Commerce** | Online share of retail |
| **Annual Retail Trade Survey** | Annual retail structure |
| **Consumer Expenditure Survey** | Household spending by demographic |

## 7.2 Housing & Real Estate

| Dataset | Use Cases |
| --- | --- |
| **Zillow Research Data** | ZHVI, ZORI, rental and housing indicators |
| **Redfin Data Center** | Weekly prices, inventory, DOM |
| **Realtor.com Data Library** | Inventory, listing price, hotness |
| **Case-Shiller / FRED** | National / metro HPI |
| **FHFA HPI** | Conforming-loan house prices |
| **Census housing starts / permits** | New construction cycle |
| **Census new home sales** | Builder demand / backlog proxy |
| **AirDNA** | Short-term rental occupancy / ADR |

## 7.3 Restaurants & Food Service

| Dataset | Use Cases |
| --- | --- |
| **OpenTable State of the Industry** | Dining recovery / seated diners |
| **Census MRTS — Food Services** | Monthly restaurant sales benchmark |

| Dataset | Use Cases |
| --- | --- |
| **Yelp Economic Average** | Business openings/closures, local demand |

## 7.4 Auto

| Dataset | Use Cases |
| --- | --- |
| **BEA auto sales** | Light vehicle sales |
| **NHTSA complaints & recalls** | Safety trends, recall pressure |
| **Ward's Auto summaries** | Sales and production context |
| **FuelEconomy.gov** | Model-level fuel economy and vehicle specs |
| **DOE Alternative Fuel Stations** | Charging/refueling infrastructure |

## 7.5 Travel, Lodging & Leisure

| Dataset | Use Cases |
| --- | --- |
| **STR / CoStar Benchmark** | Hotel occupancy and RevPAR |
| **Inside Airbnb** | STR listings and city-level market structure |
| **NPS Visitation** | Recreation and tourism demand |
| **Las Vegas visitor statistics** | Gaming, convention, visitation |

## 7.6 Gaming & Digital Entertainment

| Dataset | Use Cases |
| --- | --- |
| **Steam Charts** | Concurrent player counts |

| Dataset | Use Cases |
| --- | --- |
| SteamSpy | Ownership estimates and rankings |

## 7.7 CPG & Grocery

| Dataset | Use Cases |
| --- | --- |
| USDA ERS Food Price Outlook | Food inflation forecasts |
| **BLS CPI — Food at Home** | Grocery inflation |
| USDA AMS Market News | Wholesale produce / meat pricing |

# 8. TMT — Technology, Media, Telecom

## 8.1 Semiconductors

| Dataset | Use Cases |
| --- | --- |
| WSTS | Semiconductor sales by region |
| **SIA market data** | Chip sales commentary and regional stats |
| **SEMI Book-to-Bill** | Former semi-equipment cycle series |
| **CHIPS Act tracker / public docs** | Domestic fab awards and policy tracking |
| **US Census semiconductor shipments** | Manufacturing shipments |

## 8.2 Software, Developer Ecosystem & AI Builders

| Dataset | Use Cases |
| --- | --- |
| GitHub REST API docs | Stars, forks, commits, repo activity |
| PyPI Stats | Package download trends |
| npmtrends | JS package popularity |
| Hugging Face Model Hub | Models, datasets, downloads, leaderboards |
| Stack Overflow Developer Survey | Language / framework trends |
| OpenRouter rankings | Relative traffic/ranking across models in OpenRouter ecosystem |
| Papers with Code | Research momentum, benchmarks |

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## 8.3 Internet & Digital Advertising

| Dataset | Use Cases |
| --- | --- |
| **IAB Internet Advertising Revenue Report** | Digital ad spend by format/device |
| **Comscore public insights** | Audience and traffic commentary |
| Similarweb | Web traffic and engagement |

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## 8.4 Streaming & Media

| Dataset | Use Cases |
| --- | --- |
| Nielsen Top 10 | Weekly top streaming titles |
| **Box Office Mojo** | Film performance |
| **The Numbers** | Media / video economics |
| Spotify Charts | Music streaming demand |

## 8.5 Telecom & Connectivity

| Dataset | Use Cases |
|---|---|
| **FCC National Broadband Map** | Provider availability and ISP competition |
| **FCC BDC downloads** | Provider-reported availability data |
| **FCC Funding Map** | Subsidy / funding overlays |
| **NTIA BEAD dashboard** | BEAD program progress |
| **FCC Auctions** | Spectrum and auction data |
| **Ookla Global Index** | ISP / country speed benchmarks |

## 8.6 Cloud & AI Infrastructure

| Dataset | Use Cases |
|---|---|
| **Hugging Face Open LLM Leaderboard** | Historical model benchmark reference |
| **MLCommons MLPerf** | Hardware training/inference benchmarks |
| **Epoch AI** | Training compute, models, clusters, AI infra trend data |
| **Stanford HAI AI Index** | Annual AI adoption / investment / policy trends |